

2022-2023 计组期末试卷（回忆版）

一、填空

1. For the _____ access memory and _____ access memory, the time to access a given location is independent of the sequence of prior accesses and is constant.
2. The basis for the performance advantage of a two-level memory is a principle known as _____. This principle states that memory references tend to cluster.
3. Each instruction cycle is made up for shorter subcycles (e.g. fetch, _____, _____, _____) The execution of each subcycle involves one or more shorter operations called _____, which are the functional, or atomic, operations of a processor.
4. The term _____ parallelism refers to the degree to which, on average, the instructions of a program can be executed in parallel. _____ parallelism is a measure of the ability of the processor to take advantage of the above parallelism.
5. Register window scheme provides an efficient organization for storing _____ variables in registers. However, this scheme does not address the need to store _____ variables, those accessed by more than one procedure.
6. A very simple form of addressing is _____ addressing, in which the address field contains the effective address of the operand. With _____ addressing, the address field refer to the address of a word in memory, which in turn contains a full-length address of the operand.
7. A variety of approaches have been taken for dealing with conditional branches, please list any three approaches: (1) _____ (2) _____ (3) _____.
8. A computer has a cache and a main memory. If a referenced word is in the cache, 10ns is required to access it. Otherwise, a two-level access is performed, in which an additional 100ns is needed to load the block with the word into the cache. If the cache hit ratio is 0.9s, the average time required to access a reference word is _____ ns.
9. The basic tasks of a microprogrammed control unit are microinstruction _____ and microinstruction _____.
10. A processor has 300 different instruction operations and 70 general purpose registers. An instruction of 32-bit length has an opcode, two register operand fields and an immediate operand field. The number of bits available for the immediate operand field is _____ at most.

二、翻译

1. If there is user-visible stack addressing, then typically there is a dedicated register that points to the top of the stack. This allows implicit addressing; that is, push, pop, and other stack instructions need not contain an explicit stack operand.
2. If variable-length instructions are used, then another sort of procedural dependency arises. Because the length of and particular instruction is not known, it must be at least partially decoded before the following instruction can be fetched. This prevents the simultaneous fetching required in a superscalar pipeline.
3. Processors make use of instruction pipeline to speed up execution. In essence, pipelining involves breaking up the instruction cycle into a number of separate stages that occur in sequence, such as fetch instruction decode instruction, determine operand addresses, fetch operands, execute instruction, and write operand result.

4.The instruction format defines the layout fields in the instruction. Instruction format design is a complex undertaking, including such consideration as instruction length, fixed or variable length, number of bits assigned to opcode and each operand reference, and how addressing mode is determined.

三、选择

- 1.To maintain cache coherency, possible approaches include the following items except ()
A. Bus watching with write through
B. Hardware transparency
C. Unified caches
D. Noncacheable memory
- 2.Which of the following is not the input of the control unit ()
A. General-purpose register
B. Clock
C. Instruction register
D. Control signals from control bus
- 3.Which is the fastest memory among the following items? ()
A. Secondary
B. Main
C. Cache
D. Auxiliary Memory
- 4.In executing a microprogram, the address of the next microinstruction to be executed is in one of these categories except ()
A. Determined by instruction register
B. Determined by program counter
C. Branch
D. Next sequential address
- 5.A PC-relative mode branch instruction is 3 bytes long. The address of the instruction, in decimal, is 256029. The branch target address is (), when the signed displacement in the instruction is -12.
A. 256017
B. 256018
C. 256019
D. 256020

四、简答

- 1.List the characteristics of RISA.
- 2.分析图表题 忘了
- 3.Please briefly describe the data flow during the fetch cycle.

五、Problem

1. Consider a cache with a line size of 32 bytes and a main memory that requires 30ns to transfer a 4- byte word. What is the average number of times that the line must be written before being swapped out for a write-back cache to be more efficient than a write through cache?

备注：Ch4 题目，这是部分还是全部忘记了

2. Consider the following loop for a RISC machine.

LD R1, #0	;R1 = 0
LP LD R2, Memory (R1)	;R2 = Memory(R1)
ADD R3, R3, R2	;R3 = R3+R2
ADD R1, R1, #1	;R1 = R1+1
BLE R1, #100, LP	;Branch to LP, if R1<100

Showing the compiled program using these mentioned techniques. (loop unrolling, delayed branch and delayed load)

3.(1) Convert the following formula from infix to reverse Polish:

$X = A * B - C * D + E$

(2) Write programs to compute the above formula using stack.

4. CPU 大题，ADD 指令，具体的忘了，附上下方要求：(starting from a high address and progressing to a lower one, and the stack pointer(SP) points to the top item in the stack)

5. Given the following instruction sequence:

I1: $R2 \leftarrow 5$

I2: $R3 \leftarrow (R1) + (R2)$

I3: $R2 \leftarrow (R2) - 1$

I4: $R1 \leftarrow (R1) + (R2)$

a. Identify the write-read, write-write, and read-write dependencies in the instruction sequence.

b. Use register renaming method to improve the independency of the sequence.

c. Identify dependencies of the instruction sequence after register renaming.

6. Ch21 大题，具体的忘了